# **Henry Geerlings** | Resume

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*PhD candidate seeking full-time physical metallurgical/materials engineering position. Experience in materials characterization, foundry metallurgy, and data processing tools.*

## Education

**Colorado School of Mines** — **Golden, CO**
*PhD Materials Science* — *2021 - 2025*

**Colorado School of Mines** — **Golden, CO**
*M.S. Materials Science* — *2016 - 2018*

**University of California** — **Berkeley, CA**
*B.S. Materials Science & Engineering* — *2011 - 2015*

## Employment

**Colorado School of Mines** — **Golden, CO**
*PhD Candidate in Center for Advanced Non-Ferrous Structural Alloys (CANFSA)* — *Aug. 2021 - June 2025*
Investigating the effects of residual copper on weldability and mechanical properties of low carbon steel sheet and plate.
Detailed Achievements:
- Performed thermodynamic simulations to guide alloy/anneal design and predict laser weld hot-cracking susceptibility.
- Quantified differences in annealing, welding, and tensile behavior of steel sheet and plate as a function of copper content.
- Measured hot-cracking and cross-weld tensile properties for spot and laser welds. Characterized weld microstructures and fracture surfaces using optical and scanning electron microscopy.
- Foundry TA for sand and investment casting events, as well as the MTGN300L, "Foundry Metallurgy Lab" course.

**CoorsTek** — **Golden, CO**
*Materials Data Engineer in CoorsTek Center for Advanced Materials (CCAM)* — *Aug. 2018 - July 2021*
Lead efforts within the Computational Materials group to assess and adopt best practices within R&D ceramics data.
Detailed Achievements:
- Standardized data structure among multiple R&D programs in order to effectively share and reach target properties more effectively. Primarily investigated thermomechanical behavior of alumina and silicon nitride material systems.
- Developed and deployed high-throughput data processing tools to extract various features from historical datasets.
- Used Python and SQL to create and maintain various data pipelines for data fetching, processing, and modeling.

**Colorado School of Mines** — **Golden, CO**
*Researcher in Alliance for Development of Additive Process Technologies (ADAPT)* — *Feb. 2016 - June 2018*
Defect and powder characterization in laser-based powder bed fusion additive processes.
Detailed Achievements:
- Performed 3D $\mu$X-ray CT analysis of additively manufactured Inconel 718 samples and alloy powder stocks.
- Developed high-throughput routines for extracting, analyzing, and correlating porosity CT data with additive processing parameters and mechanical properties.
- Defined shape descriptors to quantify and compare porosity and powder particle "shapes" using image processing tools.

## Publications & Conferences

· H. Geerlings, J. Klemm-Toole, A. Clarke, K. Clarke, **Effects of residual copper on the laser weldability of steel sheet**, *AWS Professional Program, FabTech Orlando,* 2024.
· H. Geerlings, J. Klemm-Toole, A. Clarke, K. Clarke, **Weldability of Scrap Plate and Sheet with Residual Copper**, *MS&T Conference, Columbus,* 2023.
· H. Geerlings, **TRACR: a software pipeline for high-throughput materials image analysis–an additive manufacturing study**, *Colorado School of Mines Thesis Repository*, 2018.
· M. Poschmann, J. Lin, H. Geerlings, I. Winter, D.C. Chrzan, **Strain-induced variant selection in heterogeneous nucleation of $\alpha$-Ti at screw dislocations in $\beta$-Ti**, *Phys. Rev. Materials 2*. 2018.
· B. Kappes, S. Moorthy, H. Geerlings, D. Drake, A. Stebner, **Machine learning to optimize additive manufacturing parameters for laser powder bed fusion of Inconel 718**, *9th International Symposium on Superalloy 718 and Derivatives*, 2017.
· M. De Jong, W. Chen, H. Geerlings, M. Asta, K. Persson, **A database to enable discovery and design of piezoelectric materials**, *Scientific Data 2*, 2015.

## Computing

**OS**: Windows, macOS, Ubuntu (Linux)
**Technical**: Lammps, Excel, JMP, ParaView
**Utility**: Git, Azure DevOps, MS Office
**Languages**: Python, Bash, Matlab, SQL, LaTeX

## Equipment

**Metallography**: LOM, SEM, EBSD, EDS
**Thermal**: Gleeble, Dilatometry, Box Annealing
**Mechanical**: Tensile, High Cycle Fatigue, Charpy
**Joining**: Arc Weld Robot, Laser/Spot Welding, $\sigma$-Jig

## Coursework and Miscellaneous

**Materials Science and Engineering**:

Crystallography, Bonding, and Defects
Phase Transformations and Kinetics
Mechanical Behavior of Materials
Experimental Materials Science
Thermodynamics in Materials
Strengthening Mechanisms
Materials Characterization
Forging and Forming
Solidification
Corrosion

**Mechanical Engineering and Other**:

Simulation of Advanced Manufacturing Processes
Engineering Analysis using FEM
Continuum Mechanics and Dynamics
Fatigue and Fracture
Computational Linear Algebra
Heat Transfer